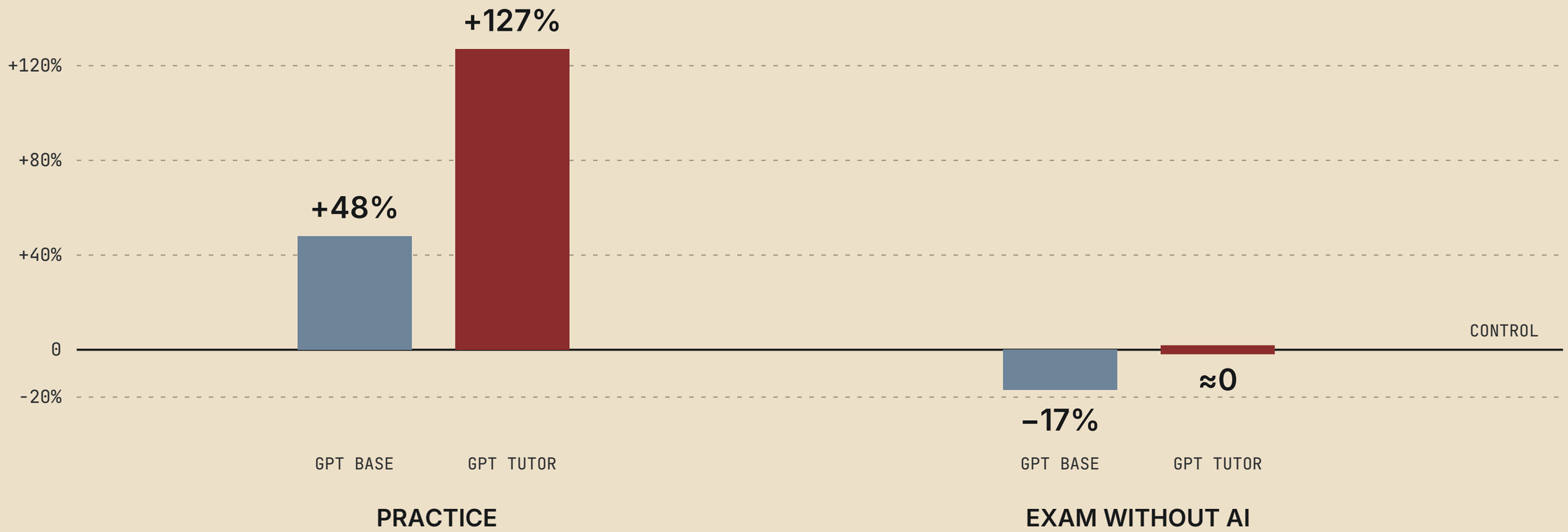
SHADOW 06 / FIGURE 9

Practice versus exam



Unrestricted access improved practice scores by 48% and lowered exam scores by 17% once access was removed. A tutor designed not to hand out answers removed the loss.

Fan et al. (BJET) — ChatGPT group improved essay scores most; knowledge gain and transfer did not differ (D08).

SOURCE Bastani et al. · Generative AI without guardrails can harm learning: Evidence from high school mathematics · PNAS 122(26), 2025 · D01–D02; companion D08.